

Angel Aldaco

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Arts in Computer Science, Political Science, Minor in Journalism

2025

- Relevant Coursework: Structure and Interpretations of Computer Programs, Data Structures, Designing Information Devices and Systems, Discrete Mathematics and Probability Theory, Effective Algorithms and Intractable Problems, Foundations of Data Science, Great Ideas of Computer Architecture (Machine Structures), Web Development, User Interface Design and Development, Operating Systems and Systems Programming, Digital Design and Integrated Circuits, Computer Architecture and Engineering, Computer Security, Database Systems, Introduction to Robotics, Designing STAR Assessments Tools, Introduction to American Politics, Political Psychology and Involvement, Introduction to Comparative Politics, History of Early Modern Political Theory, Introduction to Empirical Analysis and Quantitative Methods, Public Opinion Voting and Participation, Immigrants and Immigration as U.S. History, American Public Policy, California Politics, Public Economics, Constitutional Law of the United States, The American Presidency, The Scientific Study of International Conflict, Algorithms Public Policy and Ethics, Media Ethics, Introduction to News Reporting

EXPERIENCE

Undergraduate Student Instructor

May 2022 – Present

University of California Berkeley

Berkeley, California

Computer Security (CS 161) Head Teaching Assistant

May 2024 - Aug 2024

- Co-Developed Course Policy, Gave Lectures on Network Security Unit, Managed Staff, Wrote Exams, Ran Exam Logistics, Held Office Hours, Taught Discussions, Held Office Hours, And Did Design Reviews With Students for Projects such as the End to End Encrypted File System (see projects section for more details)

Data Structures and Algorithms (CS61B) Teaching Assistant

January 2023 - May 2024

- Taught Discussions, Labs, Held Office Hours, and Held Support Meetings for Course of up to 1600 Students
- Held Topical and Exam Review Sessions, Wrote and Graded Exam Questions, Produced Videos, Held Small Group Tutoring Sessions, Co-Authoring the Course Textbook, Mentored Academic Interns Through Pedagogical Training on a Weekly Basis, Contributed to Curriculum Design and Staff Organization, and Reviewed Project Designs for Students Through Projects Such as a Data Processing and Data Structure Management such as NGordNet (see projects down below for details)

Structures and Interpretation of Computer Programs(CS61A) Teaching Assistant

May 2022 - August 2022

- Taught Sections, Held Office Hours, Created Video Content, and Held Support Meetings for the Introductory Computer Science Course
- Made Video Content to Guide Students through Homeworks and Projects that included Creating an OOP-based game, and building a Scheme-based Interpreter

Student Advocates' Office Case Worker

August 2024 – Present

Associated Students of the University of California

Berkeley, California

Academic Case Worker

August 2024 - Present

- Advocated for and Provided Guidance to Students When They Had an Open Case Regarding Academic Issues with the University Including But Not Limited To: Grade Appeals, Degree Requirement Appeals, Following Up on Academic Misconduct Procedures, and Academic Withdrawals
- Participated in Policy Groups to Analyze University Policy in Order For the Office to Recommend Policy Changes to the University. Policies Include Reaching Out to Professors In Order To Create a Monthly Newsletter Addressed To Professors On Students' Rights and Procedures Along With Office Policy Updates and Analyzing the Working Conditions of Residential Assistants in the University.

PROJECTS

- Pintos** | *C, System Design* August 2022 – December 2022
- Designed, Programmed, and Implemented an Operating System with a team of 4, Pintos including Handling User Programs, Argument Passing, System Calls, Strict Priority Scheduling for Threads, Dealing with User Threads, Buffer Caches, and Sub Directories
 - Within Pintos Implemented Lists Structures for Kernels, a Functioning Shell with Operations Such As Pipelining, HTTP, and Implementing Memory Management
- NGordNet** | *Java, Data Parsing and Processing* April 2023
- Parses Data CSVs on the History Of Words to Build an Interactive Data Base of That History in order to Build a Graph of Hyponyms and Hypernyms of Words, along with an Efficient Algorithm for Obtaining The Greatest Common Ancestor Between Words
- End to End Encrypted File System** | *Go, System Design, Computer Security* July 2023 – August 2023
- Designed and Implemented an Encrypted File System Where User Could: (1) Create Users, (2) Login with Correct Information, (3) Create Files, (4) Append to the File at a Constant Time, (5) Share Files to Other Users to Append and Read Files, (6) Revoke Access To File That was Previously Shared
 - This Defended Against 2 Adversaries, One which Could See All of the Contents of the Database, And Another Which was a Revoked User Trying to Re-Gain Access to the File And Created Tests for such Adversaries
- RookieDB** | *Java, Database Management, SQL, MongoDB, NoSQL* August 2023 – December 2023
- Implemented a DataBase System Including Implementing B+ Tree Indices, Join Algorithms, Query Optimizations, Concurrency with Queuing and Multigranularity Locks, and Recovery with the ARIES Recovery Algorithm. To Prepared For the Project, SQL Queries were coded up, and After the Project NoSQL entries (Using MongoDB) were Coded up.
- Prairie Learn Element** | *Python, JavaScript, HTML, CSS, Mustache, Design, Docker* January 2024 – May 2024
- Using PraireLearn's Pre-existing Code Base, Designing a New Element To More Efficiently Test Students Understanding of Sorting Algorithm Iteration per Iteration Through a Drag and Drop Mechanism with Randomized Elements in the List Along with Running a Study for it's Educational Use
- NumC** | *C, Program Optimization* April 2022 - May 2022
- A Version of NumPy Written for C with Matrix Operations implemented for C. These are Spedup with SIMD instructions, OpenMP, Algorithmic Optimizations (such as Repeated Squaring), Reducing Function Calls, Loop Unrolling, and Cache Blocking
- Gitlet** | *Java, Data Structures, Algorithms* May 2023 - June 2023
- A Miniaturized Local Version of Git Which Mimics Git Behavior for init, add, commit, rm, log, global-log, find, status, restore, branch, switch, rm-branch, reset, and merge. This Requires A Clean System Design (Done Through an Updated Design Document) Which Implements Persistence Along with Maintaining Contents, Along with Implementing Graph Algorithms.
- Nonholonomic Control** | *Python, NumPy, ROS, Controls, Applied Math, Path Planning* February 2024 - March 2024
- Explained and Analyzed Path Planning Methods on a Bicycle Model. Algorithms Include: Sinusoidal Planning, Rapidly-Exploring Random Trees Planning, and Optimization Based Planning.
 - Following Implementation, This was Then Tested on an Obstacle Courses for All Planners Demonstrated Different Case-Scenarios One Would Use One Planner Over Another.
- State Estimation** | *Python, NumPy, ROS, Applied Math* March 2024
- Analyzed and Implemented Classical State Estimation Models Including: Dead Reckoning, Kalman Filtering, and Extended Kalman Filtering.
 - Following Implementation of Them, They Were Then Tested on Unicycle Model and a Quadcopter Model.
- Grasp Planning with Sawyers** | *Python, NumPy, ROS, Applied Math* April 2024
- Analyzed and Implemented Classical State Estimation Models Including: Dead Reckoning, Kalman Filtering, and Extended Kalman Filtering.
 - Following Implementation of Them, They Were Then Tested on Unicycle Model and a Quadcopter Model.
- Discretized Model of a Soft Gripper** | *Python, NumPy, ROS, Applied Math, Soft Robotics* April 2024 - May 2024
- Modeled and Implemented a Discretized Model of a Soft Gripper as Infinite Links on an Open Chain Manipulator. This is Often Done Using Bezier Curves Which We Did Model, But The Novel Aspect of The Project is Modeling It With Hermite Curves and Limit Analysis.
 - Following Theory Analysis, and Implementation The Code Was Tested On Soft Grippers to See How Practical It Was In Comparison to Our Model.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, Go, Verilog, R, RISC-V, Chisel

Developer Tools: Git, Docker, VS Code, Visual Studio, ROS, IntelliJ, MongoDB

Libraries: pandas, NumPy, Matplotlib